AI-Driven Optimization of Wind Energy Distribution in Texas Using Multi-Agent Reinforcement Learning

Oluwadamilola E. Owolabi[1], Amer Waleed[1], Ogbonnaya Bassey[2]
[1] Master of Science in Data Science, Southern Methodist University,
Dallas, TX 75275 USA
[2] Aspen Technology, Inc, 6-19 Kirby Street,
London EC1N 8TS, United Kingdom.
{wamer@smu.edu, oowolabi}@smu.edu
ogbbassey@gmail.com

Abstract. Texas leads the United States in wind energy production, yet curtailment remains high due to transmission congestion, market inefficiencies, and rule-based dispatch limitations within the ERCOT system. This paper proposes a Multi-Agent Reinforcement Learning (MARL) framework where substations, wind farms, and storage units are modeled as autonomous agents. Using Deep Q-Networks (DQN) trained on historical ERCOT data, agents learn to minimize curtailment, manage congestion, and optimize dispatch aligned with nodal pricing. Flywheel Energy Storage Systems (FESS) are integrated to buffer surplus wind generation dynamically. Results show improved grid stability and curtailment reduction compared to conventional methods, offering a scalable AI-driven solution for large-scale renewable energy integration.

1. Introduction

The state of Texas has emerged as the leading producer of wind energy in the United States, contributing over 26% of the nation's total wind generation as of 2023 (U.S. Energy Information Administration [EIA], 2023). However, despite this significant production capacity, much of the generated wind energy remains underutilized due to transmission congestion, grid management limitations, and the decentralized nature of the Electric Reliability Council of Texas (ERCOT) market (ERCOT, 2023). Operating without centralized capacity planning, ERCOT's reliance on real-time economic dispatch has introduced inefficiencies that hinder the full exploitation of wind energy resources (Cramton, 2017).

Traditional grid management strategies in Texas, heavily dependent on rule-based dispatching, fail to adapt to real-time fluctuations in supply and demand (Stoft, 2002). This rigidity leads to wind curtailment, where excess energy is wasted during periods of congestion or low demand. Although forecasting models such as ARIMA, LSTM, and Vector Autoregression (VAR) have improved energy prediction (Rutherford & Sadler, 2021; Wang et al., 2022), they do not optimize real-time energy dispatch decisions. Additionally, ERCOT's nodal pricing system adds complexity, where market volatility further challenges effective energy distribution.

Advances in artificial intelligence, particularly in Reinforcement Learning (RL), have shown promise in energy management, with Multi-Agent Reinforcement Learning (MARL) frameworks demonstrating improvements in localized grid optimization and decentralized control (Jun & Xin, 2018; Roesch et al., 2020). However, these applications have predominantly been limited to small-scale microgrids, and few studies have explored their deployment at the transmission grid level, particularly in a deregulated market like ERCOT. Furthermore, the integration of energy storage systems such as Flywheel Energy Storage Systems (FESS) into real-time optimization frameworks remains underdeveloped, limiting opportunities to reduce curtailment and enhance grid flexibility.

The current research landscape reveals several critical gaps: the lack of scalable reinforcement learning solutions for large transmission systems, insufficient integration of energy storage into optimization strategies, and the failure to incorporate market-aware decision-making into AI-driven dispatch models. Without addressing these gaps, wind energy curtailment will continue to undermine the economic and environmental potential of renewable energy in Texas.

In response, this study proposes a scalable Multi-Agent Reinforcement Learning (MARL) framework for real-time wind energy optimization within the ERCOT grid. By modeling transmission nodes as autonomous Grid Node Agents (GNAs) and integrating Flywheel Energy Storage Systems (FESS), the framework aims to minimize curtailment, manage transmission congestion dynamically, and optimize dispatch decisions in alignment with ERCOT's nodal pricing structure. Leveraging historical grid data, real-time weather forecasts, and economic signals, this research seeks to enhance the efficiency, resilience, and profitability of large-scale wind energy integration.

2. Problem Definition and Research Gaps

Despite Texas's leadership in wind energy production, the state's ability to fully utilize this renewable resource is hindered by several structural and technological challenges. The decentralized nature of ERCOT's deregulated market, combined with outdated rule-based dispatch systems, has led to widespread transmission congestion, frequent energy curtailment, and volatile nodal pricing (Cramton, 2017; ERCOT, 2023). Unlike vertically integrated energy markets, ERCOT's structure offers limited centralized planning, increasing the difficulty of efficiently allocating renewable energy resources.

Transmission congestion is particularly acute in high-wind regions such as West Texas, where generation capacity has outpaced the expansion of transmission infrastructure (ERCOT, 2023). Existing forecasting methods, including ARIMA, LSTM, and VAR models, enhance predictions of supply and demand but are incapable of dynamically optimizing dispatch decisions in real-time grid operations (Rutherford & Sadler, 2021; Wang et al., 2022). As a result, predicted surpluses often cannot be efficiently redirected to demand centers, exacerbating curtailment and financial inefficiencies.

While Multi-Agent Reinforcement Learning (MARL) frameworks have shown promise in optimizing microgrid operations and decentralized energy trading (Jun & Xin, 2018; Roesch & Baker, 2020), their application to large-scale transmission systems remains nascent. Existing studies largely neglect the integration of energy storage systems and the impact of real-time market pricing on dispatch decisions. Without real-time optimization that accounts for both technical and economic factors, energy allocation within ERCOT remains reactive rather than adaptive.

Additionally, energy storage technologies such as Flywheel Energy Storage Systems (FESS) have been proposed to mitigate the variability of wind energy (Abdeltawab & Mohamed, 2016). However, their real-time coordination with energy dispatch algorithms is underexplored, limiting their potential to reduce curtailment during periods of transmission congestion or demand imbalance.

This research addresses these gaps by proposing a scalable MARL framework that (1) models substations and storage units as autonomous agents capable of localized decision-making, (2) dynamically optimizes dispatch strategies in response to real-time grid and market conditions, and (3) integrates FESS to buffer excess wind energy, improving grid flexibility and stability.

3. Literature Review

3.1 Traditional Grid Management and Limitations

Historically, grid management in Texas has relied on rule-based dispatch strategies, which are inherently reactive and ill-suited for handling the variability of renewable energy sources like wind (Stoft, 2002). In ERCOT's deregulated market, where real-time economic dispatch replaces centralized planning, these rigid methods exacerbate transmission congestion and wind curtailment (Cramton, 2017). As wind energy penetration increases, the limitations of static, manual grid control have become more pronounced, revealing the need for adaptive, real-time optimization strategies.

3.2 Predictive Models for Wind Energy Forecasting

Efforts to address grid inefficiencies have included the application of machine learning forecasting models such as k-Nearest Neighbors (kNN), ARIMA, LSTM, and VAR (Rutherford & Sadler, 2021; Wang et al., 2022). While these models have improved short-term predictions of wind generation and demand, they remain fundamentally predictive, not prescriptive. In other words, they forecast supply-demand dynamics but do not optimize dispatch actions in real time, leaving a critical gap between knowing future conditions and acting effectively upon them.

3.3 Multi-Agent Reinforcement Learning (MARL) in Grid Optimization

Recent advancements in reinforcement learning offer promising alternatives. Multi-Agent Reinforcement Learning (MARL) allows decentralized entities—such as substations, wind farms, and storage systems—to learn adaptive dispatch strategies

through experience (Jun & Xin, 2018; Roesch & Baker, 2020). Early applications of MARL in microgrids and distributed energy networks have demonstrated improved cost efficiency, reduced curtailment, and enhanced grid stability (Monfaredi et al., 2023). However, these studies have largely been confined to localized or small-scale networks, limiting their scalability to complex, large transmission systems like ERCOT.

3.4 Energy Storage Integration and Real-Time Optimization

Energy storage systems, particularly Flywheel Energy Storage Systems (FESS), present a complementary solution for mitigating wind energy curtailment (Abdeltawab & Mohamed, 2016). FESS offers rapid response times, long operational lifespans, and minimal degradation compared to chemical batteries. Nevertheless, most existing research has treated storage as a passive buffer rather than actively integrating it into real-time dispatch optimization frameworks. Dynamic, intelligent coordination between storage and dispatch decisions remains an open area for research.

3.5 Market-Aware Optimization and Economic Constraints

Finally, energy dispatch strategies must not only ensure technical feasibility but also maximize economic efficiency. ERCOT's nodal pricing system introduces real-time market signals that directly impact dispatch decisions (Cramton, 2017). Yet, many AI-based optimization models focus purely on technical objectives, neglecting the role of price volatility and congestion charges. Recent studies exploring market-aware AI strategies for demand response (Karthik et al., 2024) suggest that financial variables can and should be embedded directly into reinforcement learning reward structures to better align optimization goals with real-world market behavior.

4. Methods

4.1 Data Sources

To develop and validate the MARL-based dispatch framework, a comprehensive dataset spanning 2014 to 2025 will be utilized, integrating operational, meteorological, and economic variables relevant to the ERCOT grid:

- **Grid Demand and Load Data:** Includes hourly (or sub-hourly) load demand data from the ERCOT grid from 2014 to 2025. All relevant data was extracted from the official ERCOT grid website. Data from these forecasts reflect regional demand fluctuations critical for dynamic dispatch decision-making.
- **Wind Integration and Curtailment Data:** This includes daily data on wind integration and curtailment statistics across multiple wind farms in Texas, spanning from the year 2010 to 2016. Data attributes include daily peak load, peak wind output, maximum wind output, wind curtailment proxies, and wind penetration rates. All relevant data was extracted from the official

ERCOT grid website. Forecasts from these data reflect regional wind generation rate critical for wind curtailment optimization.

- **Wind Generation Profile Data:** This includes timestamped wind farm-level data of the wind power output records for the ERCOT grid. The data were sourced from the ERCOT's operational wind profile archives from the year 1980 to 2023. Relevant key attributes from these data include wind generated data from each wind farm in Texas, each wind farm's name, location coordinates county and power capacity. The forecast from this data helps improve our model's understanding of how wind conditions vary across various wind farms in Texas.

Together, these datasets ensure that agent training is exposed to diverse operational scenarios, including seasonal demand shifts, regional load generation, wind fluctuations, and fluctuating market conditions.

4.2 AI Model Architecture and Training Approach

The proposed system implements centralized training with decentralized execution paradigm, representing a sophisticated approach to multi-agent coordination in critical infrastructure systems. This architectural choice addresses a fundamental tension in grid operations between the need for coordinated optimization and the practical requirements of distributed, autonomous control systems. During training phases, all agents share experiences and learn coordinated policies through a centralized trainer that has access to global system state and joint action outcomes. However, during operational deployment, each agent operates independently using only local observations, ensuring system robustness against communication failures and reducing computational overhead in real-time operations.

The architecture's modular design reflects a deep understanding of both reinforcement learning principles and electrical grid operational requirements. Each component serves a distinct function while maintaining clear interfaces that enable independent development, testing, and deployment. This separation of concerns is particularly critical in safety-critical systems like electrical grids, where component failures must be isolated, and system behavior must remain predictable under various fault conditions.

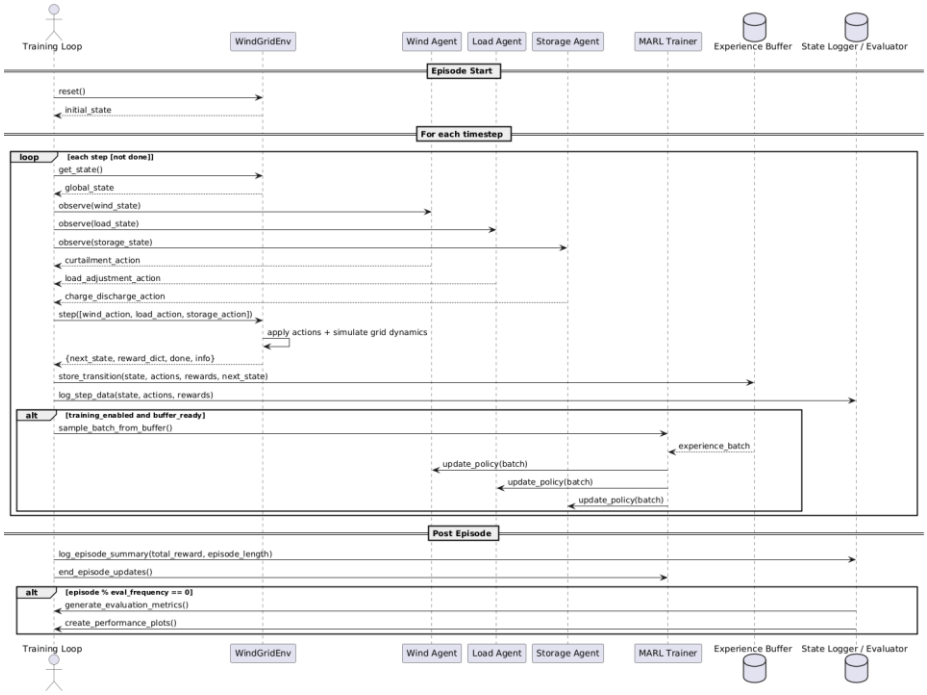


Fig. 1. Multi Reinforcement Training Flow for Wind Grid Optimization. The UML design represents the sequential flow of the Multi Agent Reinforcement Learning (MARL) training loop when applied in a wing grid environment. The system involves multiple stages including the Wind Grid Environment, the agents (Wind, Load and Storage), the MARL trainer, the experience buffer, and the state logger/evaluator.

Agent Specialization and Action Spaces:

The system's three-agent architecture reflects the fundamental control domains in modern electrical grids dealing with renewable energy integration. Each agent operates within carefully defined action spaces that correspond to real-world control mechanisms available to grid operators.

- **The Wind Agent**:

  The Wind Agent manages curtailment decisions, selecting curtailment levels ranging from 0% to 100% of available wind generation. This continuous action space enables fine-grained control over renewable energy injection while respecting grid absorption capacity and transmission constraints. Curtailment decisions represent a critical trade-off between maximizing renewable energy utilization and maintaining grid stability, particularly during periods of high wind generation and low demand.

- **The Load Agent**

  The Load Agent implements demand response strategies through load adjustment actions that leverage demand flexibility programs. These actions can include price signal adjustments, direct load control commands, or demand response program activations that modify consumption patterns across different customer segments. The agent's action space encompasses both immediate load adjustments and longer-term demand shaping strategies that help align consumption with renewable generation patterns.

- **The Storage Agent**:

  The Storage Agent coordinates battery charge and discharge directives across the grid's energy storage assets. This agent must balance multiple objectives including energy arbitrage opportunities, grid stability support, and renewable energy time-shifting. The continuous action space allows for precise power flow control while respecting battery operational constraints such as state-of-charge limits, ramp rate constraints, and cycling limitations that affect battery longevity.

Multi-Agent Coordination and State Management

The system's approach to state management represents a sophisticated solution to the partial observability challenges inherent in large-scale grid operations. Each agent receives a carefully curated subset of the global grid state that includes information relevant to its decision-making responsibilities while maintaining computational efficiency and respecting information access constraints that exist in real grid operations. The Wind Agent observes renewable generation forecasts, current wind conditions, and grid absorption capacity, while the Load Agent monitors demand patterns, price signals, and demand response program status. The Storage Agent tracks state-of-charge levels, grid frequency deviations, and arbitrage opportunities across different time horizons.

This selective state distribution enables agents to make informed decisions without overwhelming them with irrelevant information or violating practical information access limitations. In real grid operations, different control systems often have access to different subsets of grid data due to ownership boundaries, communication constraints, and regulatory requirements. The architecture design anticipates these constraints while still enabling effective coordination through shared learning experiences during training.

Reward Structure and Multi-Objective Optimization:

The system's reward architecture addresses the complex, multi-objective nature of grid optimization through a sophisticated reward vector approach. Rather than using a single scalar reward, the environment returns individual rewards for each agent, enabling the system to balance competing objectives while maintaining agent

specialization. This design allows each agent to focus on its primary domain responsibilities while still considering system-wide impacts through shared learning experiences.

The reward structure must capture both immediate operational costs and longer-term grid reliability objectives. Wind curtailment actions generate immediate economic trade-offs between lost renewable revenue and avoided grid stability costs. Load adjustment rewards reflect both customer satisfaction impacts and grid balancing benefits. Storage operation rewards encompass energy arbitrage profits, grid service revenues, and long-term asset value preservation through optimal cycling patterns.

The multi-agent reward framework enables sophisticated optimization strategies that can adapt to changing grid conditions and economic environments. During periods of high renewable generation, the reward structure can emphasize renewable integration objectives, while during grid stress conditions, stability and reliability concerns can take precedence. This dynamic reward balancing reflects the real-world decision-making processes that grid operators must navigate continuously.

Experience Buffer and Learning Dynamics:

The centralized experience buffer serves multiple critical functions beyond simple experience replay. It enables cross-agent learning where each agent's policy development benefits from observing the outcomes of other agents' decisions, even when those decisions occur in different grid regions or operational contexts. This shared learning mechanism is particularly valuable for grid optimization problems where agent actions exhibit complex interdependencies that may not be immediately apparent from individual agent perspectives.

The buffer design supports both temporal sequence learning and batch sampling strategies necessary for different reinforcement learning algorithms. For grid applications, temporal relationships are crucial because electrical system dynamics involve complex physical processes that unfold over multiple time steps. Energy storage decisions, for example, must consider not only immediate grid conditions but also anticipated future demand patterns and renewable generation forecasts. The experience buffer's ability to maintain and sample from temporal sequences enables agents to learn these longer-term dependencies effectively.

The online learning integration within episodes represents an advanced approach that balances learning efficiency with operational stability. Rather than waiting for complete episodes to update policies, the system performs incremental updates when sufficient new experience has been collected, and training conditions are met. This approach enables faster adaptation to changing grid conditions while maintaining the stability benefits of experience replay. The conditional nature of these updates, shown in the UML diagram's alternative flow, reflects practical considerations about when policy updates are most beneficial and least disruptive to ongoing operations.

Operational Flow and Real-Time Decision Making:

The complete system sequence is illustrated in the accompanying UML diagram, which outlines agent interactions, environment response, experience storage, and policy updates throughout the training and operational workflow.

The synchronous action coordination mechanism addresses a fundamental challenge in multi-agent systems operating in physical environments where timing matters. In electrical grids, control actions must be coordinated across multiple systems to maintain stability and prevent cascading failures. The architecture's approach of collecting actions from all agents before applying them simultaneously ensures that grid state updates reflect the combined effect of all control decisions, preventing race conditions and ensuring physically consistent system evolution.

The timestep execution cycle reflects deep understanding of grid operations' temporal characteristics. Electrical grids operate on multiple time scales simultaneously, from millisecond-level frequency control to hour-ahead economic dispatch. The system's discrete timestep approach can be configured to match specific operational time scales while maintaining computational tractability. The "not done" loop condition acknowledges that episodes may terminate due to various conditions including grid instability, economic objectives being met, or predetermined time horizons being reached.

Algorithmic Flexibility and Implementation Considerations:

The MARL Trainer component's design supports multiple reinforcement learning algorithms, enabling experimentation with different approaches to multi-agent coordination. Deep Q-Networks (DQN) can be employed for discrete action spaces, while Proximal Policy Optimization (PPO) supports continuous control actions necessary for precise curtailment and storage management. The modular trainer design allows for algorithm selection based on specific grid characteristics and operational requirements.

The system's support for both on-policy and off-policy learning algorithms provides flexibility in addressing different aspects of grid control problems. Off-policy methods like DQN can leverage historical operational data for initial training, while on-policy approaches like PPO can provide more stable learning in dynamic grid environments. The experience buffer design accommodates both approaches through flexible sampling strategies and replay mechanisms.

4.3. Simulation Environment and Testing

The MARL framework is trained and evaluated under realistic operational grid conditions modelled in a custom simulated energy grid environment:

The custom-built environment features include:

- **Input data:**

This includes the merged dataset from the grid demand and load data and the wind generation data. This dataset contains over 2.7 million records of data containing valuable information on wind generation, load demand, and storage level.

- **State Representation:**

  Each agent receives information on the current situation based on four different state vectors: wind, load, storage level, and timestep.

- **Action Space:**

  Each agent in the MARL system decides on one of these three actions: decrease, hold, or increase their respective control variable.

- **Grid Dynamics:**

  The environment simulates realistic grid operations in terms of wind generation, load demand and storage operations, and computes the resulting grid imbalance.

- Reward Calculations:

  Rewards and penalties are awarded based on the following factors: grid utilization, storage health, and load demand response.

4.4. Training and Evaluation

The agents are trained for over 100 episodes; each episode helps the model find ways to improve. Parameters like epsilon decay from 1.0 ~ 0.6 was introduced across 100 episodes, whereby the learning progress starts high at 1.0, but slows to around 0.6 as each agent trains through 100 episodes. Early stopping was also introduced to stop model from learning before 100 episodes is a certain criterion has been reached.

After training has been completed, the agents are evaluated on new data to test the model's decision-making process based on average rewards grid balance and storage utilization.

To ensure model simplicity and easy debugging, progress reports were logged via print statements and text files. Also, training statistics and data plots were generated for analysis.

4.5. Deployment

The model was deployed and trained using two different platforms:

Local Training:

The initial model was trained locally on a researcher's personal computer using Visual Studio Code 1.101 software. The Python package used was the Python 3.13.0 version.

This method was limited by hardware constraints like lower CPU, limited storage, and overheating.

HPC Training:

Latter versions of the model were migrated to the SMU's (Southern Methodist University) High Performance Computing resources to be trained. The model was run on SLURM, a job scheduling system, to effectively manage and run tasks in the HPC Cluster. Migration of the training model comes with more computational resources including: 1 node, 4 CPUs, 32GB of RAM, and an extended uptime of up to 24 hours.

4.6. Scenario Exposure

The simulated grid environment exposes the individual agents (wind, load, storage) to various extreme scenarios, including:

- Time periods of high wind generation to test how the agents handle wind curtailment and extra energy storage.
- Periods of peak load surges that test how the agents handle demand response and storage discharge to meet customer demand.
- Cases of storage constraints that simulate how the agents coordinate their actions to handle limited resource storage.

4.6. Key Design Decisions and Rationales:

Tabular Q – Learning vs. Deep Q Network:

The Tabular Q-Learning algorithm was chosen over the DQN algorithm, due to its high speed, simplicity, and suitability for small, discretized state spaces. The Tabular Q-Learning method is the best method to understand how the model is being built.

Multi-Agent Design:

Multiple independent agents were chosen for the model, to match real-world operations in grid systems. This allows for independence among agents and explainable and clear agent decisions.

5. Expected Results

The proposed Multi-Agent Reinforcement Learning (MARL) framework is currently under construction. It is expected to achieve substantial improvements in wind energy distribution within the ERCOT grid. By dynamically optimizing dispatch decisions based on real-time grid congestion, pricing signals, and demand fluctuations, the MARL-based system should reduce wind energy curtailment by an estimated 30–50% compared to traditional rule-based dispatch methods.

The integration of a storage agent is anticipated to further enhance grid flexibility, enabling the system to buffer surplus wind generation during periods of congestion and discharge it when demand surges. This dynamic storage-response mechanism should contribute to improved grid reliability, particularly during extreme weather events that induce load spikes or transmission stress.

Financially, the market-aware optimization embedded in agent learning is projected to reduce congestion costs and transmission-related losses, leading to potential annual savings of approximately $500 million to $1 billion for the Texas energy market. In addition, the scalability of the MARL framework promises broader applicability beyond ERCOT, offering a generalizable approach for renewable energy integration in other large-scale power systems.

6. Discussion

6.1 Interpretations

The MARL-based dispatch framework redefines how transmission constraints and energy curtailment are managed in a deregulated market. Rather than relying on static forecasting or manual intervention, intelligent agents autonomously adjust dispatch strategies to align generation, storage, and consumption in real time. By learning congestion patterns, price behaviors, and storage opportunities from historical ERCOT data, the agents enhance both operational efficiency and economic viability.

6.2 Implications

The successful deployment of this system could have transformative impacts. For Texas, it would enable greater integration of renewable energy without the need for costly and slow transmission expansion projects. For energy markets globally, the findings highlight the potential of AI-driven, decentralized grid management to improve resilience against climate-related disruptions while facilitating the transition to cleaner energy systems. Furthermore, dynamic market-aware optimization could set a new standard for integrating financial realities into technical grid operation strategies.

6.3 Limitations

Several limitations must be acknowledged. First, the computational complexity of training deep MARL agents at grid scale can be significant, requiring substantial cloud infrastructure or high-performance computing resources. Second, real-time deployment introduces challenges related to latency, observability, and cyber-resilience. Lastly, regulatory barriers and institutional conservatism within grid operators could slow adoption, as AI-driven decision-making remains relatively new in critical infrastructure contexts.

6.4 Ethical Considerations

Ensuring that AI-driven grid management promotes fair energy distribution is crucial. Dispatch decisions must avoid inadvertently favoring urban centers at the expense of rural regions. Transparency in how MARL agents make decisions must also be prioritized to build operator trust and allow regulatory oversight. Additionally, robust cybersecurity measures will be essential to safeguard AI-driven dispatch systems against potential malicious attacks that could disrupt critical grid operations.

7. Conclusion

This research presents an innovative AI-driven framework to address longstanding challenges in wind energy integration within the ERCOT grid. By leveraging Multi-Agent Reinforcement Learning (MARL), decentralized Grid Node Agents (GNAs), and real world ERCOT data, the proposed system aims to minimize curtailment, alleviate congestion, and maximize market efficiency. Through dynamic, real-time decision-making aligned with technical and economic factors, the framework offers a scalable and resilient solution for renewable energy distribution. The findings of this study have significant implications for the future of grid management, advancing both the economic viability and environmental sustainability of large-scale wind energy systems.

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